

Proposition: $\forall n \in \mathbb{Z}^+ \ 2 \leq n$, n is itself prime, or is a product of prime numbers.

Base case: $n = 2$. 2 is prime, so we are done.

Inductive Hypothesis: Suppose $\{2, 3, \dots, k\}$ are all either prime numbers or products of prime numbers.

Inductive Step: Show $k+1$ is prime or a product of primes.

Case I: $k+1$ is prime, so we are done.

Case II: $k+1$ is composite. Then $k+1$ has positive factors other than 1 and itself. Let γ and ϕ be two such factors. We know

$$1 \leq \gamma \leq k \quad \text{and} \quad 1 \leq \phi \leq k$$

So we also know that γ and ϕ are either prime or products of primes. Write them as

$$\gamma = (a_1 \cdot a_2 \cdot \dots \cdot a_m) \quad \phi = (b_1 \cdot b_2 \cdot \dots \cdot b_r)$$

Then $k+1 = \gamma\phi = (a_1 \cdot a_2 \cdot \dots \cdot a_m)(b_1 \cdot b_2 \cdot \dots \cdot b_r)$, which is a product of primes.

◦ $\forall n \in \mathbb{Z}^+ \ 2 \leq n$, n is prime or a product of primes

Q.E.D.